

Predicting 30-Day Hospital Readmissions

An Early-Warning System to Reduce Avoidable Diabetic Readmissions

Ikechukwu Michael Jumbo – Springboard Data Science Capstone 2 Project | UCI Diabetes Dataset (1999–2008) | May 2026

71% of high-risk patients caught pre-discharge · Built on ~100K diabetic encounters

Today's Agenda

01

The Problem & Business Case

Why 30-day readmissions cost hospitals millions annually

02

Data Overview & Wrangling

101,766 encounters — cleaned & engineered to 99,343 records

03

Exploratory Data Analysis

Who gets readmitted, when, and what drives it

04

Preprocessing & Feature Engineering

How raw data became 157 predictive signals

05

Modelling & Evaluation

Logistic Regression, Random Forest, Gradient Boosting, XGBoost

06

Key Findings & Recommendations

Actionable insights for hospital operations

07

Q&A & Open Discussion

10 minutes for feedback and questions

The Problem

Why 30-Day Readmissions Demand Immediate Attention

\$26B

Medicare readmission cost
annually (US)¹

14–23%

Diabetic patients
readmitted within
30 days²

\$15,200

Average cost per
readmission episode
(CMS, 2018)³

~27%

Median share of
readmissions that are
preventable⁴

The Core Challenge

Hospitals already hold the data needed to predict readmissions — they lack the discharge-time signal to act on it. The cost of error is deeply asymmetric: a missed readmission triggers a five-figure care episode and CMS penalty exposure, while a false flag costs only a follow-up call.

We turn existing EHR data into an at-discharge risk score — making readmission prevention proactive.

¹ CMS / General Medicine (2014): Medicare readmission cost \$26B/yr; \$17B preventable. ² Rubin et al., Clin Diabetes Endocrinol (2017): DM 30-day readmission rate 14.4–22.7%. ³ HVB Program Analysis, PMC (2022): avg. readmission cost \$15,200. ⁴ StatPearls / NCBI Bookshelf (2024): median preventable readmission share 27% across 34 studies.

Data Overview & Wrangling

UCI Diabetes Dataset — 1999 to 2008

101,766

Raw hospital encounters

50

Original variables

99,343

Records after cleaning

114

Engineered features

Missing Data

Dropped weight, A1Cresult, max_glu_serum (>80% missing). Used 'Unknown' for categorical fields to preserve information.

Target Engineering

Binary readmit_30d: readmitted within 30 days (1) vs not (0). Excluded hospice/expired discharge codes.

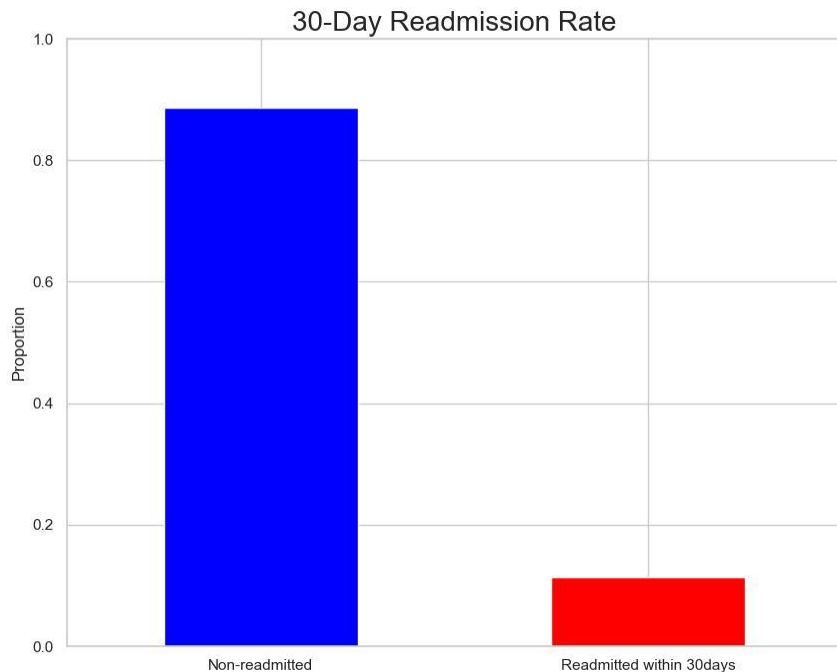
Medication Features

Each drug generated 3 signals: presence flag, ordinal dose, and dose-change instability indicator.

ICD Diagnosis Codes

Extracted 3-digit ICD-9 groups to reduce sparsity while preserving clinical category signal.

Target Variable: Class Imbalance



88.6%

Non-readmitted patients

11.4%

Readmitted within 30 days

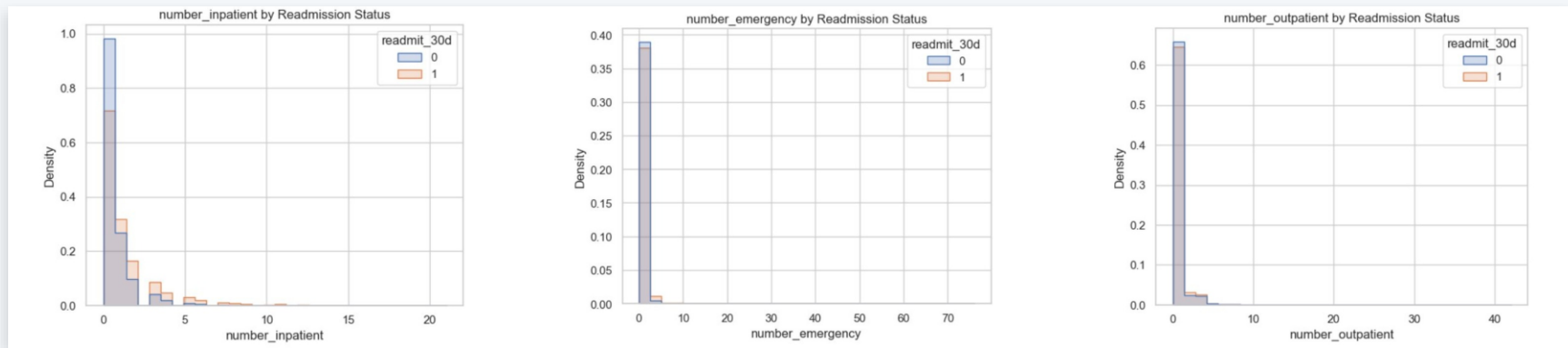
Modelling Implication

Class imbalance demands:

- Recall as primary metric
- Stratified train/test split
- Class weighting in models
- Threshold optimisation

Exploratory Data Analysis

Utilisation Distributions by Readmission Status — Where the Signal Lives



Inpatient: clear separation

Emergency: distributions overlap

Outpatient: distributions overlap

What this tells us

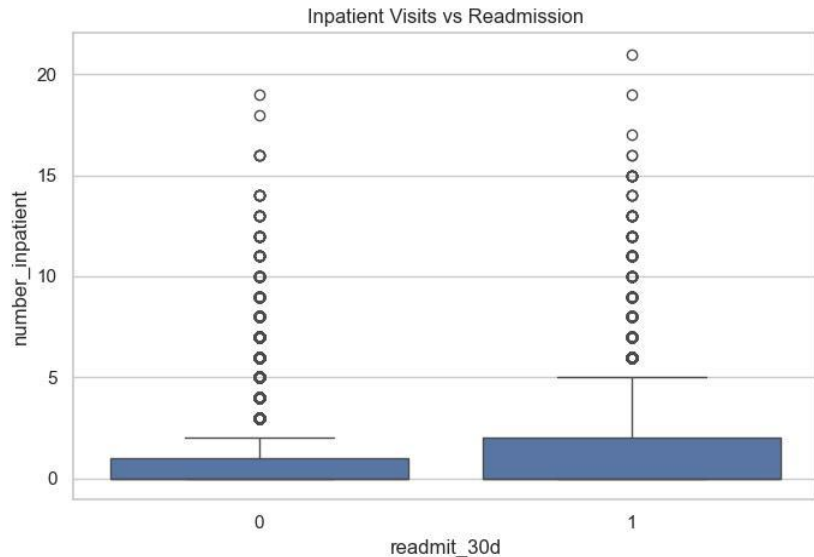
Inpatient is the only utilisation feature with a visible signal. At zero prior inpatient visits, ~98% of non-readmitted patients vs ~72% of readmitted — readmitted patients are far more likely to have a hospitalisation history.

Emergency and outpatient distributions overlap. These features carry little standalone signal — but matter when combined with inpatient history into composite utilisation features.

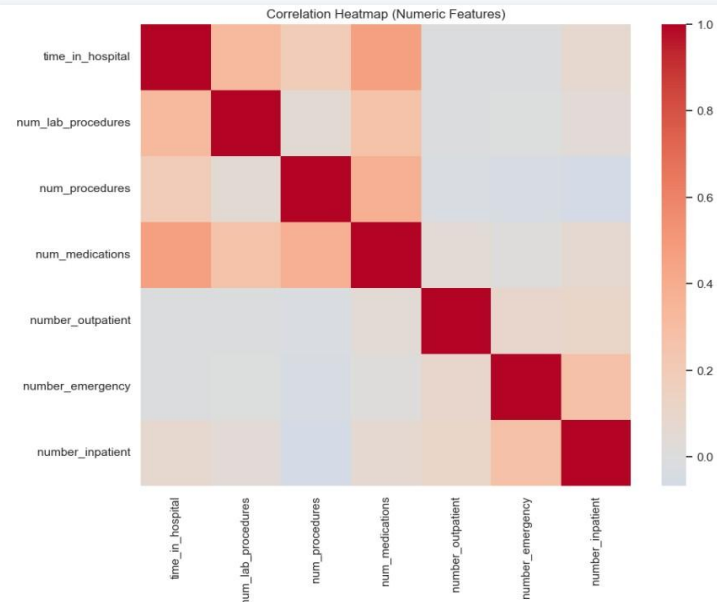
Modelling implication: build interaction features over weighted utilisation rather than relying on each variable in isolation.

Key Predictors from EDA

Inpatient Visits vs Readmission



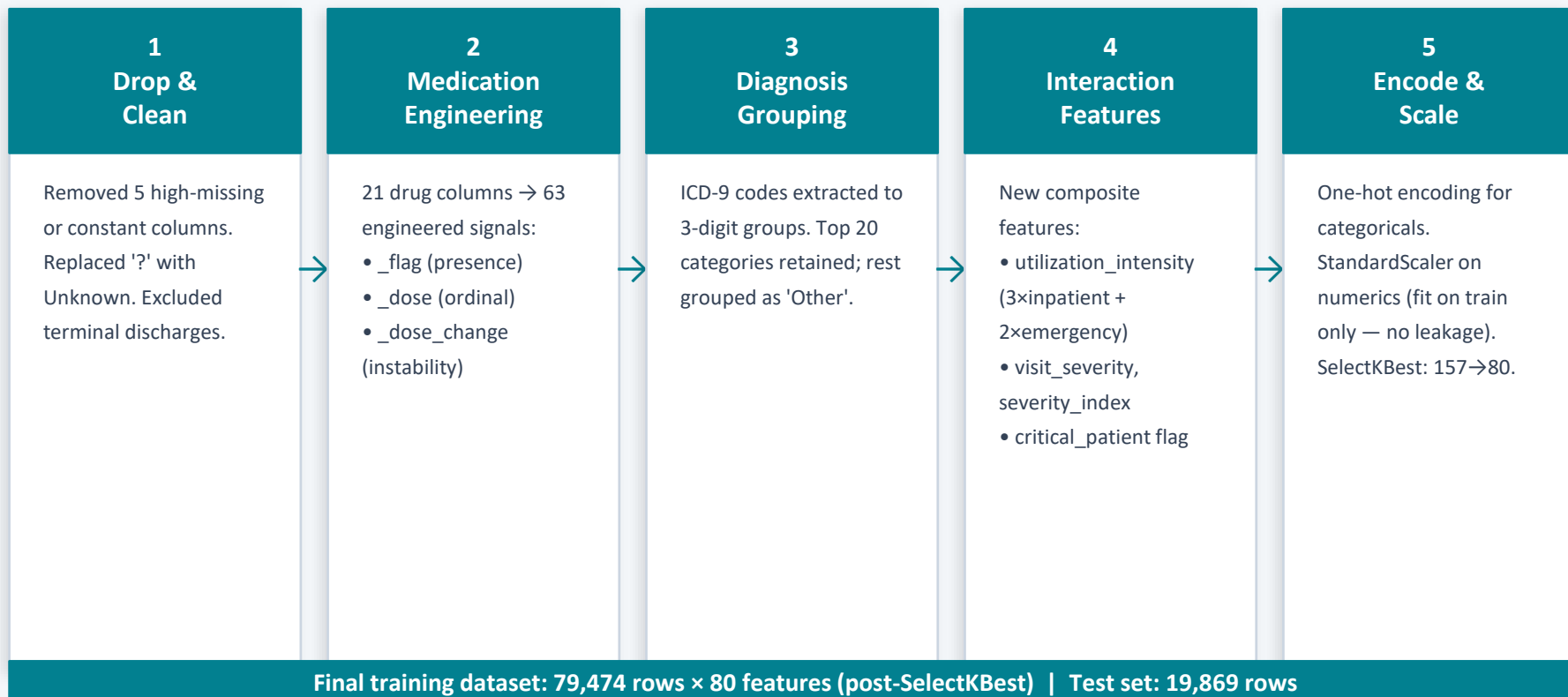
Correlation Among Numeric Features



Key Insight: number_inpatient is the single strongest individual predictor. Utilisation variables (inpatient · emergency · outpatient) show moderate positive correlation — combining them into a weighted utilisation_intensity feature amplifies the signal for tree-based models.

Feature Engineering & Preprocessing

Transforming raw clinical data into 157 predictive signals



Modelling Approach

Four models evaluated — XGBoost selected as final model

Logistic
Regression

ROC-AUC
0.644

Baseline. Interpretable, but struggles with non-linear interactions.

Random
Forest

ROC-AUC
0.613

Robust ensemble. Slightly lower performance; computationally heavier.

Gradient
Boosting

ROC-AUC
0.646

Highest raw ROC-AUC. Sequential learning on residuals.

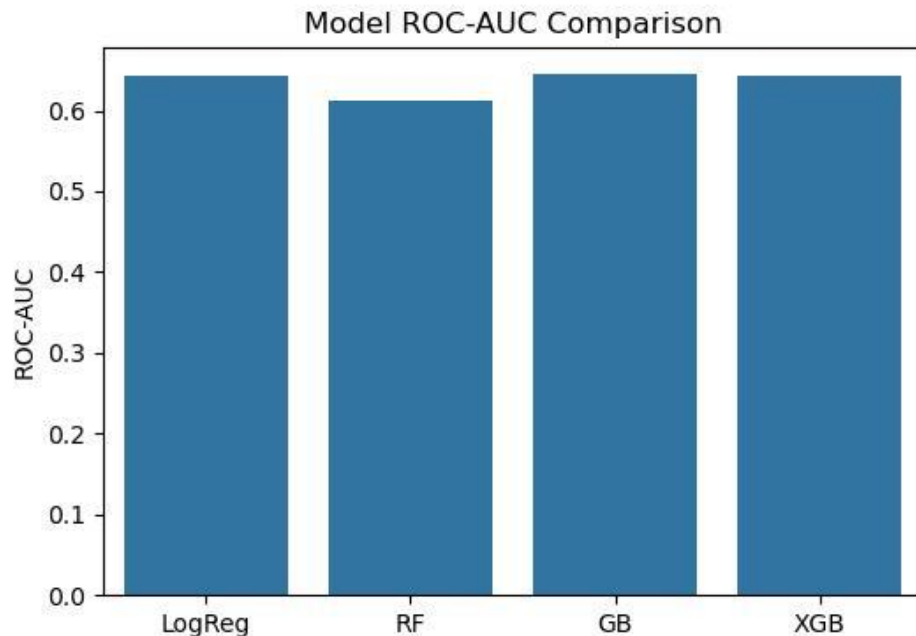
XGBoost
★ Selected

ROC-AUC
0.642

Regularisation + speed + scale_pos_weight for imbalance. Best balance of performance & flexibility.

★ XGBoost chosen for its regularisation ($\alpha + \lambda$), built-in class weighting, and best fit with interaction-heavy features.

Model Performance Comparison



~0.644 Logistic Regression

~0.613 Random Forest

~0.646 Gradient Boosting

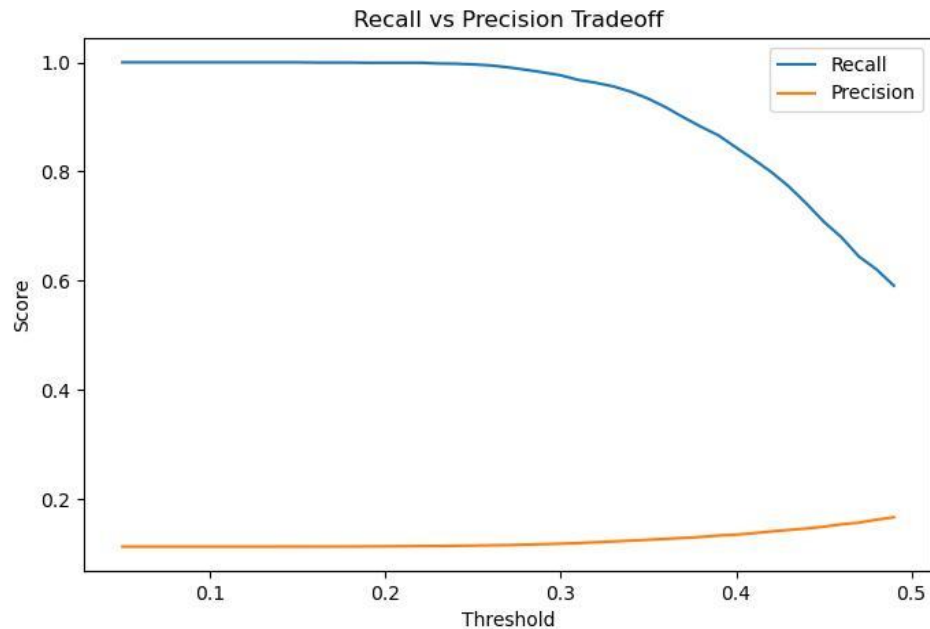
~0.642 XGBoost (selected)

Why comparable ROC-AUC?

Readmission is driven by multiple weak signals — no single feature dominates. All models plateau similarly; the differentiator is recall at the right threshold.

Threshold Optimisation

Balancing Recall and Precision to Meet Clinical Objectives



0.71

Recall — high-risk caught

0.15

Precision — flag accuracy

0.649

ROC-AUC — discrimination

0.45

Operating threshold

Why this trade-off makes business sense: a missed readmission costs the hospital ~\$15K in care plus reputational and CMS penalty exposure. A false positive costs only the marginal time of a follow-up call. The economics are deeply asymmetric — and the threshold is tuned accordingly.

Final Model Performance

Confusion Matrix & Classification Report — Test Set (19,869 patients)

Confusion Matrix

Test set: 19,869 patients · threshold = 0.45

	Predicted: No Readmit	Predicted: Readmit
Actual: No Readmit	8,547 True Negative	9,059 False Positive
Actual: Readmit	660 False Negative	1,603 True Positive

Classification Report

Per-class precision · recall · F1

Class	Precision	Recall	F1	Support
Not Readmitted	0.93	0.49	0.64	17,606
Readmitted (target)	0.15	0.71	0.25	2,263

Headline numbers

Recall 0.71 · Precision 0.15 · F1 0.25 · ROC-AUC 0.649

8,547

True Negatives — non-readmits

1,603

True Positives — high-risk caught

9,059

False Positives — extra monitoring

660

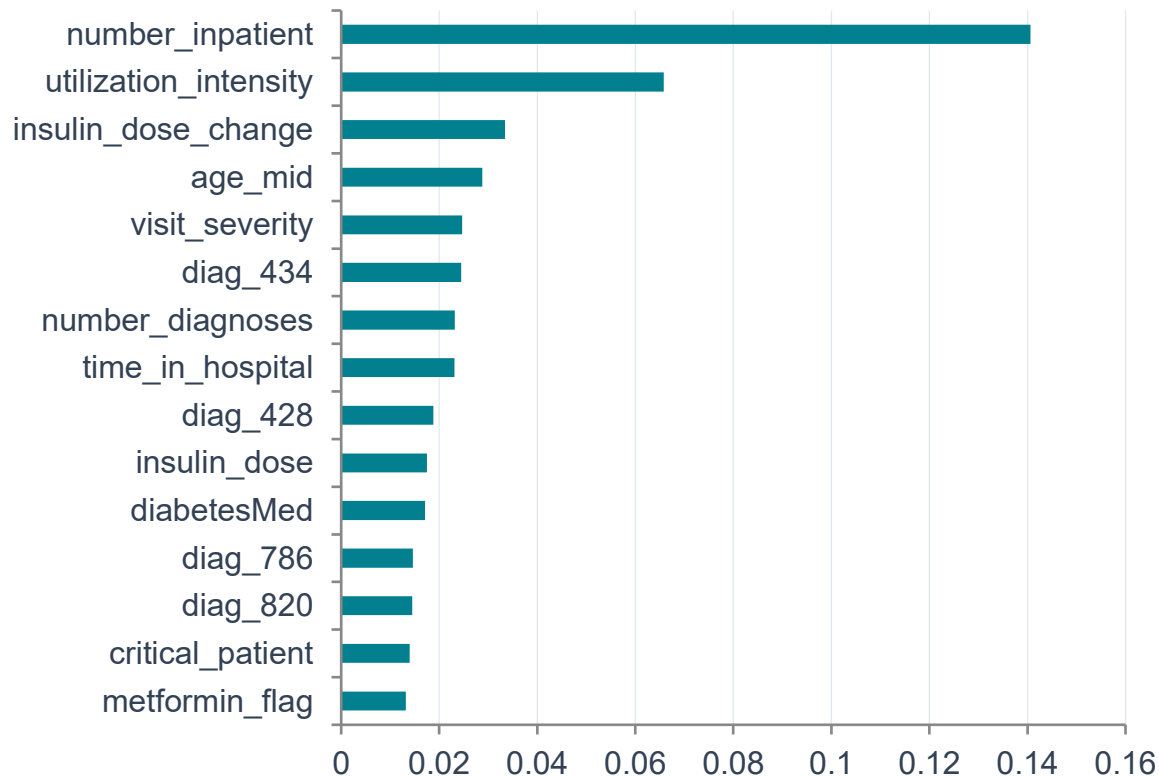
False Negatives — missed cases

Interpretation

The model captures 71 of every 100 true readmissions before the patient walks out the door. At ~\$15K per averted readmission, the captured cohort represents seven-figure annual savings at typical hospital scale.

What Drives Readmission?

Top 15 Feature Importances from the XGBoost Model



#1 number_inpatient

Prior hospitalisation is far and away the strongest single predictor of 30-day return.

#2 utilization_intensity

Engineered composite of inpatient $\times 3$ + emergency $\times 2$. Validates feature engineering value.

#3 insulin_dose_change

Medication instability signals uncontrolled diabetes — a direct clinical risk marker.

#4 age_mid

Non-linear age effect: risk is elevated at both ends of the adult age spectrum rather than in the middle.

#5 visit_severity

Interaction of total visits $\times$ hospital stay — captures the most medically complex patients.

Strategic Recommendations

Translating Model Insights into Hospital Operations

Deploy Risk Scoring at Discharge

Immediate

Embed the model in the EHR so every diabetic discharge receives a real-time risk flag. With ~7 in 10 readmissions identified pre-discharge, intervention spend finally lands where it matters.

Target Insulin-Adjusting Patients

Short-term

Insulin dose-change patients show meaningfully elevated readmission risk. A structured diabetes-nurse handover for this cohort offers the strongest dollar-per-readmission-avoided of the interventions tested.

High-Utiliser Outreach Programme

Medium-term

Patients with 2+ prior inpatient visits are the top-risk cohort. A 30-day programme of phone check-ins, medication reconciliation and dietitian referral could recover 20–30% of avoidable returns.

Expand Data Collection

Ongoing

Restoring HbA1c and BMI capture (now >80% missing) is the single largest accuracy upgrade available. Investment in clinical-data governance compounds model value over time.

Limitations & Future Work

Limitations

- Data from 1999–2008 — treatment protocols and medication landscapes have changed substantially.
- Weight, A1Cresult, and max_glu_serum were dropped due to >80% missingness — all are strong clinical predictors.
- Precision remains low (0.15) — 85% of flagged patients would NOT be readmitted. Workflow design must account for this.
- Model has not been validated on an external or prospective cohort — generalisation requires clinical validation.
- Social determinants of health (housing, income, transportation) are entirely absent from the dataset.

Future Roadmap

Prospective Validation

Deploy on current patient cohort and track model predictions against actual 30-day outcomes.

SHAP Explainability

Generate per-patient SHAP waterfall plots so clinicians understand individual risk factors.

Real-Time EHR Integration

Build an API endpoint connecting the model to live EHR data for at-discharge scoring.

Restore Clinical Features

Prioritise collection of A1Cresult and weight in ongoing patient records.

Deep Learning Exploration

Test LSTM or Transformer models on longitudinal encounter sequences for improved temporal patterns.

Summary

What We Built · Why It Matters · What We Need Next

Dataset	101,766 diabetic encounters cleaned into 99,343 records with 114 structured features
Target	Binary 30-day readmission — 11.4% positive rate (class imbalance addressed)
Best Model	XGBoost with scale_pos_weight, SelectKBest (80 features), and threshold = 0.45
Performance	Recall 0.71 · Precision 0.15 · F1 0.25 · ROC-AUC 0.649
Top Driver	number_inpatient is the #1 predictor — prior hospitalisation history is paramount
Clinical Value	71% of high-risk patients flagged before discharge — every catch is an intervention opportunity
Next Step	Pilot in one department, validate prospectively, then scale via EHR API + SHAP explanations

Questions & Open Discussion

10 minutes reserved for feedback, challenges, and ideas
